

So we will do :

TASK 1 : Students List Screen

- API: users
- Show:
 - Name
 - Email
 - City
 - Company
 - Department

TASK 2 : Announcements Screen

- API: posts
- Show:
 - Title
 - Body

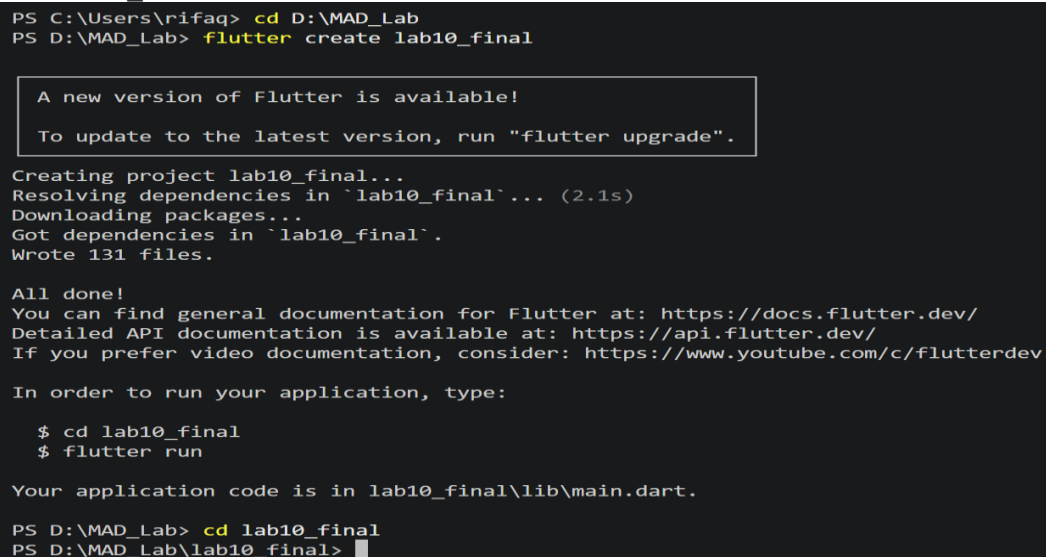
TASK 3 : Reusable Row Component

- Create ReusableRow
- Use it in UI

STEP 0 : Create Project

Open terminal and run:

```
cd D:\MAD_Lab
flutter create lab10_final
cd lab10_final
```



```
PS C:\Users\rifaq> cd D:\MAD_Lab
PS D:\MAD_Lab> flutter create lab10_final

A new version of Flutter is available!
To update to the latest version, run "flutter upgrade".

Creating project lab10_final...
Resolving dependencies in `lab10_final`... (2.1s)
Downloading packages...
Got dependencies in `lab10_final`.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

$ cd lab10_final
$ flutter run

Your application code is in lab10_final\lib\main.dart.

PS D:\MAD_Lab> cd lab10_final
PS D:\MAD_Lab\lab10_final> █
```

STEP 1 : Install HTTP

```
flutter pub add http
```

```
PS D:\MAD_Lab\lab10_final> flutter pub add http
Resolving dependencies...
Downloading packages...
+ http 1.6.0
+ http_parser 4.1.2
  matcher 0.12.19 (0.12.20 available)
  meta 1.17.0 (1.18.2 available)
  test_api 0.7.10 (0.7.12 available)
+ typed_data 1.4.0
  vector_math 2.2.0 (2.3.0 available)
+ web 1.1.1
Changed 4 dependencies!
4 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
PS D:\MAD_Lab\lab10_final> █
```

STEP 2 : Create Folder Structure

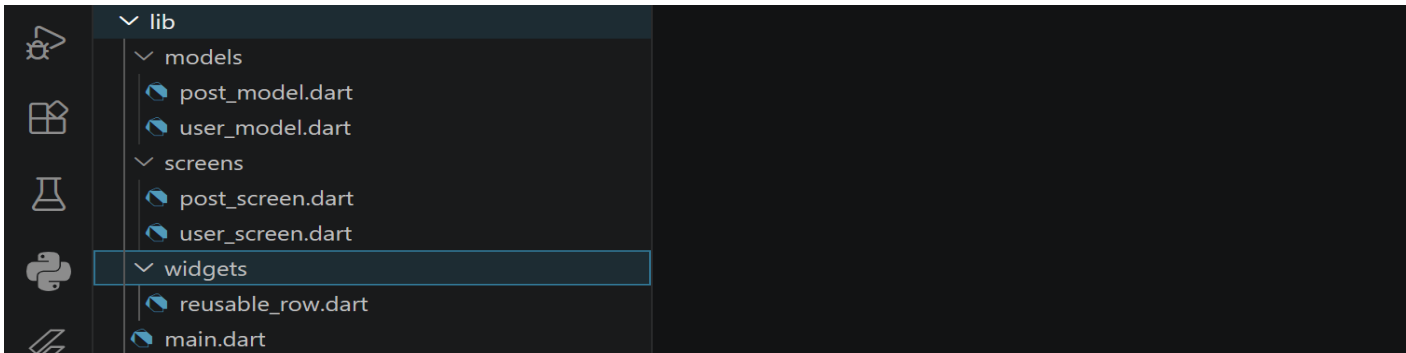
- Open folder in VS Code
- Create folders inside `lib`
- Right-click on `lib`
 - i. New Folder → `models`
 - ii. New Folder → `screens`
 - iii. New Folder → `widgets`

STEP 3 : Create Files

Now we create files inside these folders

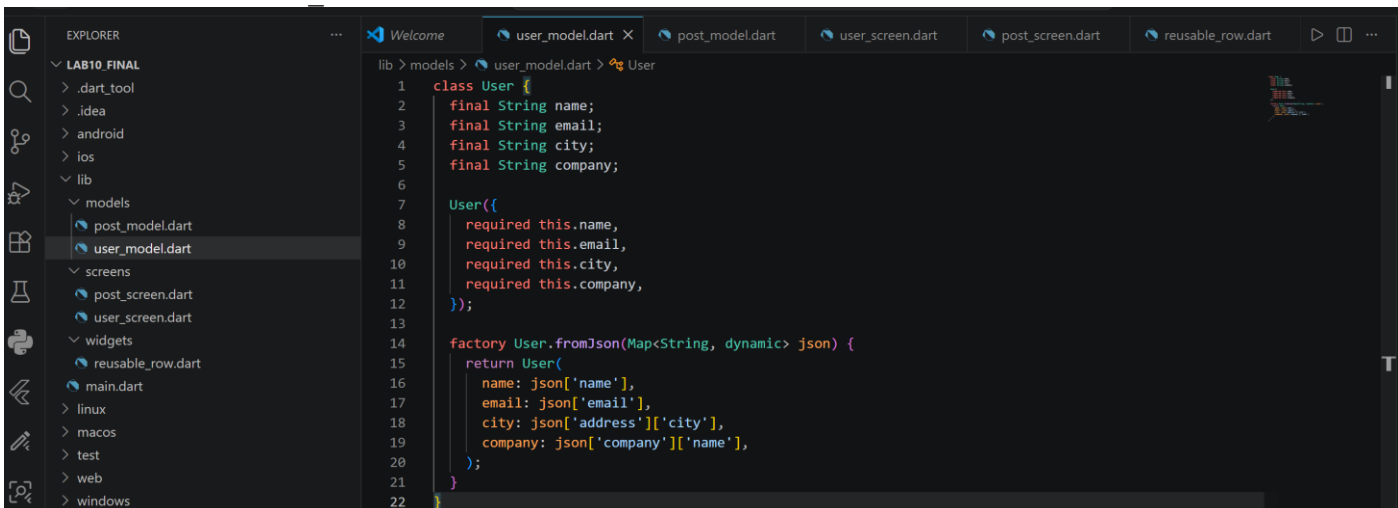
- **Inside `models`**
 - i. `user_model.dart`
 - ii. `post_model.dart`
- **Inside `screens`**
 - i. `user_screen.dart`
 - ii. `post_screen.dart`
- **Inside `widgets`**
 - i. `reusable_row.dart`

Final Structure



STEP 4 : user_model.dart

lib → models → user model.dart



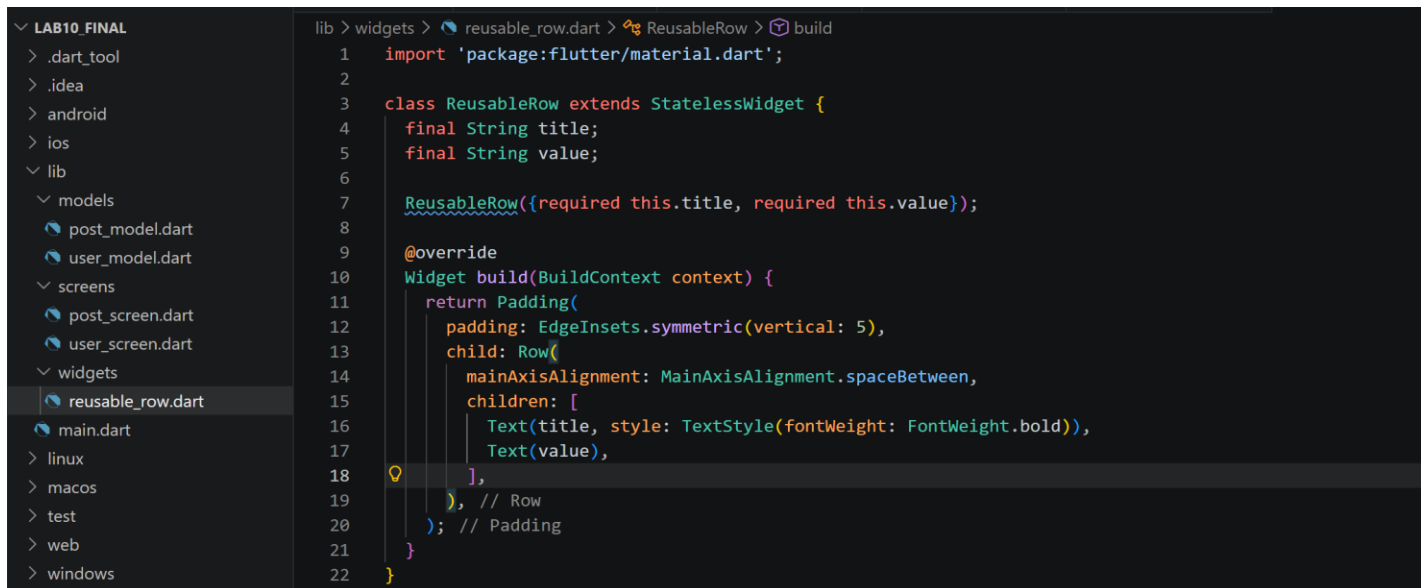
STEP 5 : post_model.dart

lib → models → post model.dart



STEP 6 : reusable_row.dart (IMPORTANT)

lib → widgets → reusable_row.dart



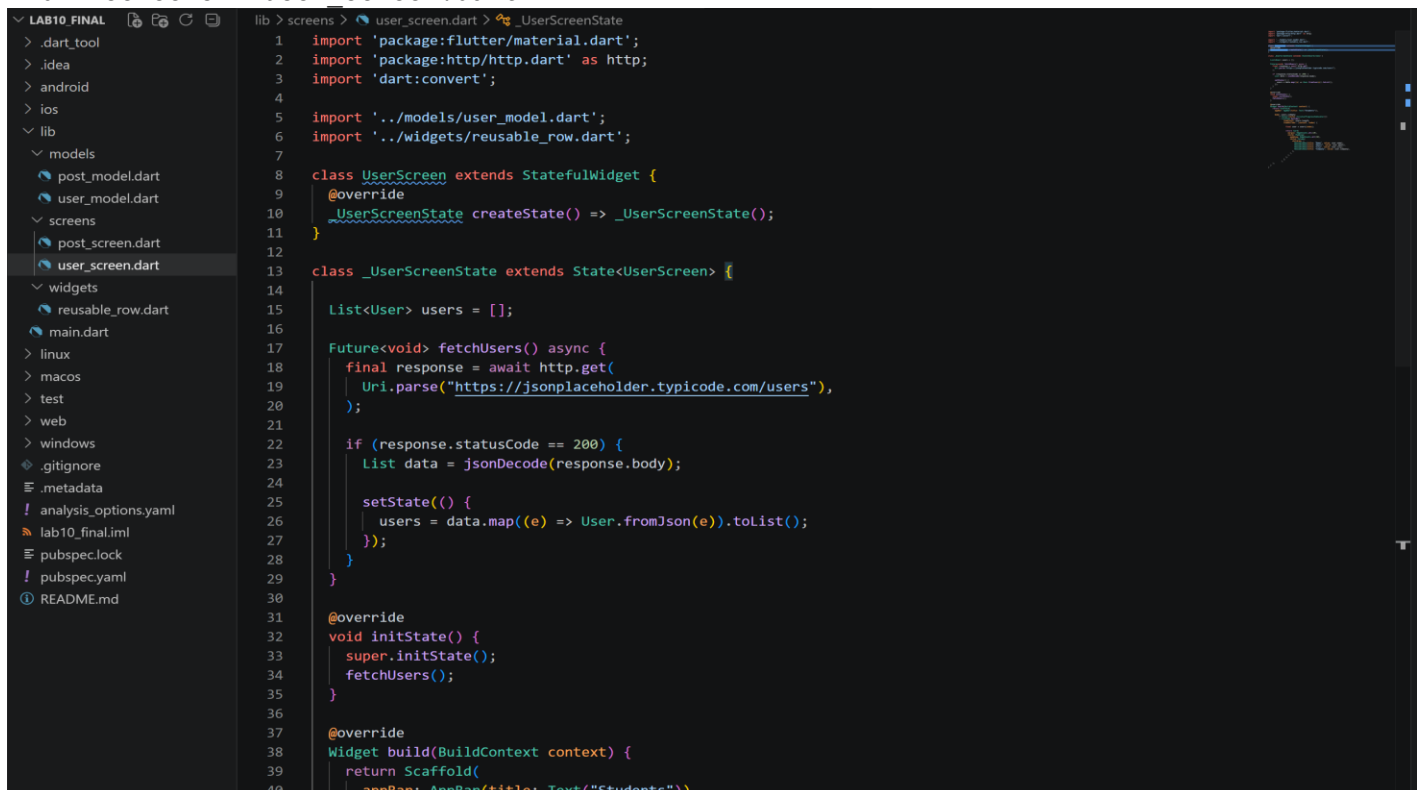
```

lib > widgets > reusable_row.dart > ReusableRow > build
1  import 'package:flutter/material.dart';
2
3  class ReusableRow extends StatelessWidget {
4    final String title;
5    final String value;
6
7    ReusableRow({required this.title, required this.value});
8
9    @override
10   Widget build(BuildContext context) {
11     return Padding(
12       padding: EdgeInsets.symmetric(vertical: 5),
13       child: Row(
14         mainAxisAlignment: MainAxisAlignment.spaceBetween,
15         children: [
16           Text(title, style: TextStyle(fontWeight: FontWeight.bold)),
17           Text(value),
18         ],
19       ), // Row
20     ); // Padding
21   }
22 }

```

STEP 7 : user_screen.dart

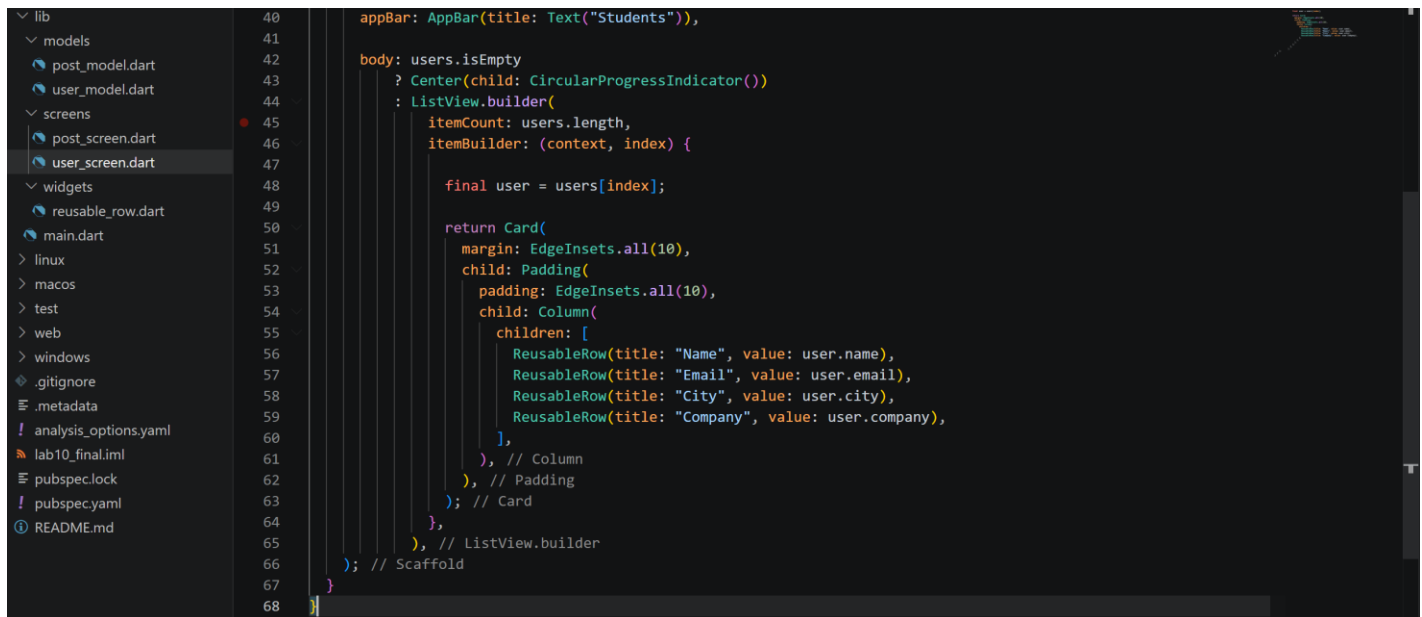
lib → screens → user screen.dart



```

lib > screens > user_screen.dart > _UserScreenState
1  import 'package:flutter/material.dart';
2  import 'package:http/http.dart' as http;
3  import 'dart:convert';
4
5  import '../models/user_model.dart';
6  import '../widgets/reusable_row.dart';
7
8  class UserScreen extends StatefulWidget {
9    @override
10   _UserScreenState createState() => _UserScreenState();
11 }
12
13 class _UserScreenState extends State<UserScreen> {
14
15   List<User> users = [];
16
17   Future<void> fetchUsers() async {
18     final response = await http.get(
19       Uri.parse("https://jsonplaceholder.typicode.com/users"),
20     );
21
22     if (response.statusCode == 200) {
23       List data = jsonDecode(response.body);
24
25       setState(() {
26         users = data.map((e) => User.fromJson(e)).toList();
27       });
28     }
29   }
30
31   @override
32   void initState() {
33     super.initState();
34     fetchUsers();
35   }
36
37   @override
38   Widget build(BuildContext context) {
39     return Scaffold(
40       appBar: AppBar(title: Text("Students")),

```



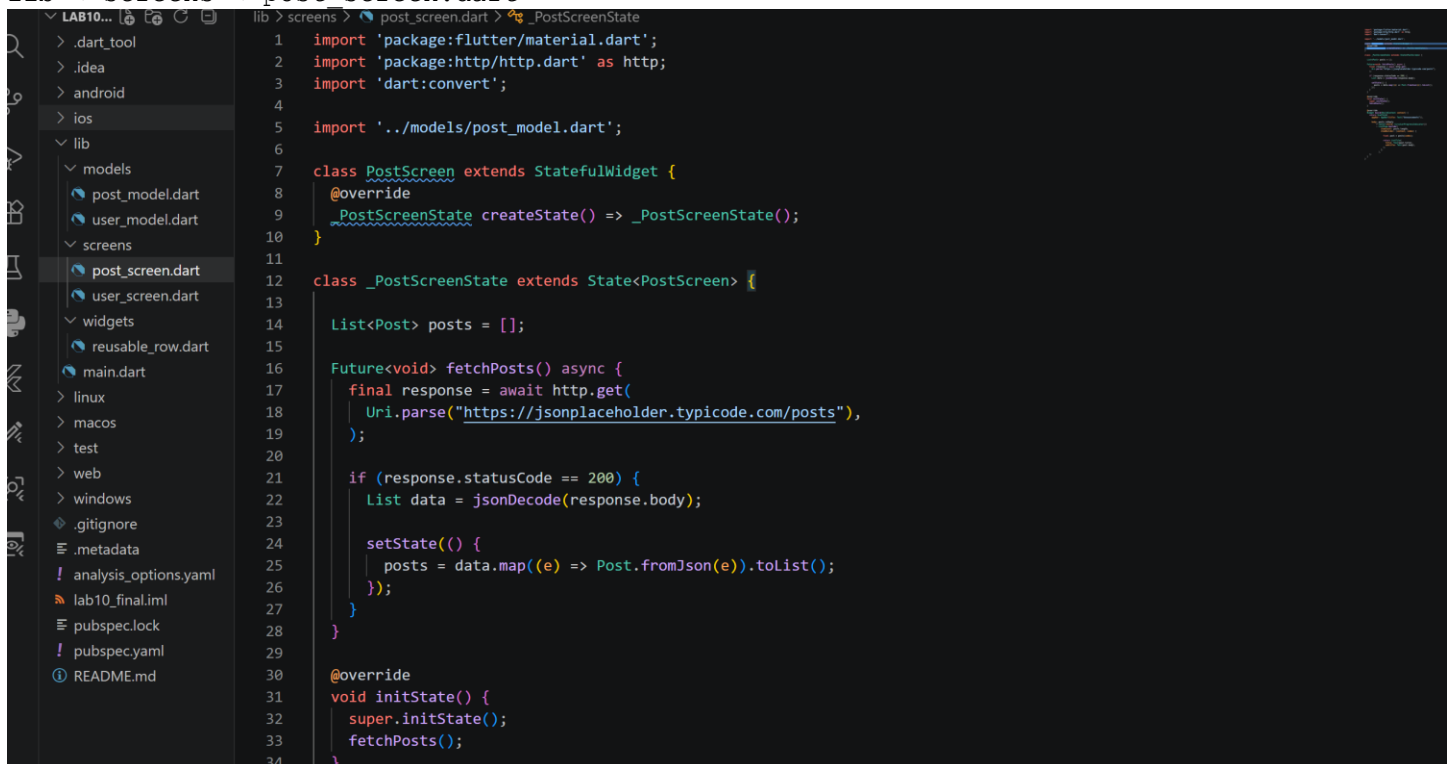
```

40      appBar: AppBar(title: Text("Students")),
41
42      body: users.isEmpty
43        ? Center(child: CircularProgressIndicator())
44        : ListView.builder(
45          itemCount: users.length,
46          itemBuilder: (context, index) {
47
48            final user = users[index];
49
50            return Card(
51              margin: EdgeInsets.all(10),
52              child: Padding(
53                padding: EdgeInsets.all(10),
54                child: Column(
55                  children: [
56                    ReusableRow(title: "Name", value: user.name),
57                    ReusableRow(title: "Email", value: user.email),
58                    ReusableRow(title: "City", value: user.city),
59                    ReusableRow(title: "Company", value: user.company),
60                  ],
61                ), // Column
62              ), // Padding
63            ); // Card
64          },
65        ), // ListView.builder
66      ); // Scaffold
67    }
68  }

```

STEP 8 : post_screen.dart

lib → screens → post screen.dart



```

1  import 'package:flutter/material.dart';
2  import 'package:http/http.dart' as http;
3  import 'dart:convert';
4
5  import '../models/post_model.dart';
6
7  class PostScreen extends StatefulWidget {
8    @override
9    _PostScreenState createState() => _PostScreenState();
10 }
11
12 class _PostScreenState extends State<PostScreen> {
13
14   List<Post> posts = [];
15
16   Future<void> fetchPosts() async {
17     final response = await http.get(
18       Uri.parse("https://jsonplaceholder.typicode.com/posts"),
19     );
20
21     if (response.statusCode == 200) {
22       List data = jsonDecode(response.body);
23
24       setState(() {
25         posts = data.map((e) => Post.fromJson(e)).toList();
26       });
27     }
28   }
29
30   @override
31   void initState() {
32     super.initState();
33     fetchPosts();
34   }

```

```

35
36
37 @override
38 Widget build(BuildContext context) {
39   return Scaffold(
40     appBar: AppBar(title: Text("Announcements")),
41
42     body: posts.isEmpty
43         ? Center(child: CircularProgressIndicator())
44         : ListView.builder(
45             itemCount: posts.length,
46             itemBuilder: (context, index) {
47
48               final post = posts[index];
49
50               return ListTile(
51                 title: Text(post.title),
52                 subtitle: Text(post.body),
53               ); // ListTile
54             }, // ListView.builder
55           ); // Scaffold
56   }
57

```

STEP 9 : main.dart

lib → main.dart

We will Delete everything and paste new Code which is:

```

1  import 'package:flutter/material.dart';
2  import 'screens/user_screen.dart';
3  import 'screens/post_screen.dart';
4
5  void main() {
6    runApp(MyApp());
7  }
8
9  class MyApp extends StatelessWidget {
10   @override
11   Widget build(BuildContext context) {
12     return MaterialApp(
13       home: HomeScreen(),
14       debugShowCheckedModeBanner: false,
15     ); // MaterialApp
16   }
17 }
18
19 class HomeScreen extends StatelessWidget {
20   @override
21   Widget build(BuildContext context) {
22     return Scaffold(
23       appBar: AppBar(title: Text('Lab 10')),
24
25       body: Column(
26         children: [
27
28           ElevatedButton(
29             onPressed: () {
30               Navigator.push(
31                 context,
32                 MaterialPageRoute(builder: (_) => UserScreen()),
33               );
34             },
35             child: Text("Students"),
36           ), // ElevatedButton
37
38           ElevatedButton(

```



```
38  
39  
40  
41  
42  
43  
44  
45  
46  
47  
48  
49  
50  
51  
52  
ElevatedButton(  
  onPressed: () {  
    Navigator.push(  
      context,  
      MaterialPageRoute(builder: (_) => PostScreen()),  
    );  
  },  
  child: Text("Announcements"),  
), // ElevatedButton  
],  
), // Column  
); // Scaffold  
}
```

FINAL STEP : RUN

In terminal:

```
flutter run -d chrome
```

Home Screen (Main Menu)

- “Students” button
- “Announcements” button



Students List Screen (Task 1)

- Name
- Email
- City
- Company

← Students	
Name	Leanne Graham
Email	Sincere@april.biz
City	Gwenborough
Company	Romaguera-Crona
Name	Ervin Howell
Email	Shanna@melissa.tv
City	Wisokyburgh
Company	Deckow-Crist
Name	Clementine Bauch
Email	Nathan@yesenia.net
City	McKenziehaven
Company	Romaguera-Jacobson
Name	Patricia Lebsack
Email	Julianne.OConner@kory.org
City	South Elvis
Company	Robel-Corkery
Name	Chelsey Dietrich
Email	Lucio_Hettinger@annie.ca
City	Roscoeview
Company	Keebler LLC

One Card Close View (Task 3)

reuable

← Students	
Name	Leanne Graham
Email	Sincere@april.biz
City	Gwenborough
Company	Romaguera-Crona

Announcements Screen (Task 2)

- Title
- Body

